

Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.
2. **Modeling.** A new “Key Assumptions” discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.
3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover’s rewards experiment shows some substitution at the margin. Appendix OF estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.
4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results (\$31 bn vs. \$27 bn).
5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 3

The assumption that consumers who hold both debit and credit cards never substitute between them is unintuitive. Even with the π probability formulation, it is unclear why this assumption is not important for results. Why is γ not enough to drive merchant adoption when debit is available as a backup? Please expand the discussion of consumer substitution (top of page 17).

Reply: γ is enough to drive merchant credit acceptance even under credit-debit substitution. The challenge is that allowing credit-debit substitution generates equilibrium predictions at odds with the empirical evidence, so the main text keeps the assumption of point-of-sale segmentation.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix OF.1). On the other hand, Discover's rewards shock (Online Appendix OD.2) shifts consumer spending onto Discover from both cash and debit, consistent with some substitution at the point of sale.

Allowing substitution generates two equilibrium predictions at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates (Online Appendix OF.2, Equation 55). Expert witnesses in *Ohio v. AmEx* instead describe merchants disciplining credit fees by threatening to route transactions to rival credit cards rather than debit, and networks report pricing credit against credit while ignoring debit. The second is that capping debit fees would lower credit fees, but the Durbin Amendment cut regulated debit interchange by roughly half without moving credit interchange (Figure 5).

Online Appendix OF estimates two alternative specifications that navigate this trade-off differently. One allows point-of-sale substitution and incremental debit sales relative to cash (Online Appendix OF.3.1) and matches the consumer-side substitution evidence most closely. The other treats debit as generating no incremental sales relative to cash (Online Appendix OF.3.2), removing the multi-homing channel from the credit-acceptance threshold so credit acceptance depends only on credit fees. It is silent on Durbin pricing because debit fees are zero by construction. For credit fee cap, merger, and dual routing counterfactuals, both alternatives deliver welfare similar to the baseline.

The treatment effects estimation relies heavily on a single grocer adopting credit cards. If this grocer engaged in advertising or a promotion, that could bias results.

Please discuss whether such events occurred, and acknowledge the limitation of relying on one case study.

Reply: Acceptance changes are rare (I find only two clear cases in the Homescan panel), and the estimates should be interpreted with this limitation in mind. That said, both events show similar patterns, suggesting the results are not driven by idiosyncratic features of any one retailer.

The main text now frames the merchant-type recovery as closer to a calibration exercise than a conventional estimation, given the limited variation in merchant adoption in a mature payments market. It also clarifies the identifying assumption and what would violate it. The key assumption is that merchants do not differentially target credit card users relative to other consumers of similar income levels. A confounding event would be one that simultaneously shifts merchant acceptance and consumer card holdings, for example a co-branded credit card campaign that onboards consumers onto a new card at the same time the merchant changes which cards it accepts. I excluded such events and verified that no similar campaigns accompanied the events I study. I verified against news reports that the acceptance changes were not accompanied by major advertising campaigns or promotions. I cannot name the retailers in the main text because the Homescan data use agreement requires retailer identities to remain anonymous.

Clarify the discussion on pages A-34 to A-35. My understanding is that consumers do not observe debit vs credit preferences at adoption (preferences γ_{it} are integrated out). If a consumer knew she had bifurcated preferences, she would adopt both cards. Please explain this assumption more clearly.

Reply: The appendix microfounds the complementarity parameter χ from the main model. The idea is that consumers draw transaction-level shocks that determine whether a given purchase suits credit or debit. These shocks are realized at the point of sale, not at adoption. At adoption, consumers integrate over the distribution of future transaction shocks when choosing which cards to carry.

You are exactly right that if a consumer knew she would have bifurcated preferences, she would want to adopt both card types. This is the case of a large variance Σ of transaction-specific preferences. This is precisely one microfoundation for why the complementarity parameter χ is positive for credit-debit pairs. Consumers who expect to sometimes prefer credit and sometimes prefer debit have strong option value from holding both. A consumer who instead knows she strongly prefers credit has less option

value from carrying debit and may benefit more from holding two credit cards, since there is also option value within a card type (e.g., different reward categories suit different purchases). The parameter χ in the main model captures this trade-off in reduced form. Empirically, $\chi^{22} = -4.82$, so the option value of a second same-type card is more than offset by adoption and management costs. I have revised Online Appendix OE.8 to make these two points explicit.

I was pleased with the passthrough discussion. Can the paper produce the counterfactuals underlying Table A.12?

Reply: Yes. Online Appendix OG.1 now re-estimates the model under zero passthrough and reports counterfactuals in Table OG.3. Under no passthrough, merchants absorb fee changes entirely within their margins rather than adjusting retail prices.

For counterfactuals with small fee and reward changes, welfare effects are close to baseline. Uncapping debit raises total welfare by \$8 billion (versus \$7 billion), dual routing yields \$2.9 billion (versus \$3.8 billion, both net of the mechanical χ^{22} adjustment described in Online Appendix OE.7), and CBDC yields \$1.4 billion (versus \$1.6 billion).

For counterfactuals with large net fee changes, deadweight loss creates differences. The fee cap welfare gain falls to \$17 billion from \$27 billion under full passthrough, and the monopoly gain falls to \$8 billion from \$15 billion. Under full passthrough, lower rewards reduce consumption while merchant price cuts stimulate it, so total consumption is roughly unchanged and deadweight loss is essentially flat; the welfare gain comes mainly from the credit-aversion channel and redistribution. Under zero passthrough, merchant prices do not adjust, so the rewards-driven cut in consumption is not offset. The cap raises deadweight loss, and total welfare gains shrink.

The surplus distribution under the fee cap also shifts: merchants retain the fee savings rather than passing them to consumers, so merchant surplus rises by \$49 billion while consumer surplus falls by \$31 billion (versus gains for both groups under full passthrough).

Table 1 should indicate which dataset it uses and the number of observations.

Reply: The updated table caption refers to the Diary of Consumer Payment Choice. I have added observation counts to Table 1 to clarify the sample sizes underlying the summary statistics.

Estimation Equation 2 does not reference recent treatment effects literature (e.g., Sun and Abraham). If these methods do not apply to triple-differences with a continuous treatment, please acknowledge explicitly.

Reply: Recent work on treatment effect heterogeneity under staggered adoption (Sun and Abraham, Callaway and Sant’Anna) addresses bias when already-treated units serve as controls. This concern does not apply here. The control group consists of grocery stores that accepted credit cards throughout the sample. The two merchants that changed acceptance are the treated units, and they are never used as controls for each other. The estimated effect is a simple average treatment effect across the two events, not a staggered difference-in-differences. I have added language to Section III.B clarifying this design.

In Section III.C.2, how is "carrying" a card measured? Homescan likely does not record this. Please clarify.

Reply: I measure “carrying” based on observed usage in the Homescan data, not self-reported ownership. A consumer is classified as carrying a card if they use it during the sample period. Among non-cash consumers, I define the primary card as the one with the most trips and the secondary card (if any) as the one with the second-most trips. This usage-based measure is appropriate for the model, which captures payment behavior rather than ownership. A credit consumer who owns a debit card but never uses it is effectively a credit-only user for merchant acceptance decisions.

Why does P^w have a superscript "w"? Does the price index depend on a consumer’s wallet, or is it being distinguished from another P ?

Reply: Yes, the price index depends on the consumer’s wallet. I have clarified this in the text following Equation 6: “ P^w is a CES price index that depends on other merchants’ actions and on the consumer’s wallet.” Intuitively, a consumer carrying Visa sees a lower effective price at Visa-accepting merchants than a cash-only consumer does, so P^w aggregates prices weighted by how much utility a consumer with wallet w derives from each merchant’s acceptance decisions.

In the displayed equation near the bottom of page 21, should y be indexed by i ?

Reply: I appreciate the suggestion. The notation suppresses the i subscript on y to avoid cluttering the equations. Throughout the model section, y represents a generic consumer's income rather than being indexed by i . The context makes clear that y refers to the income of the consumer under consideration. I have opted to keep the notation as is to improve readability.

The acronym PCE should be spelled out the first time it appears.

Reply: Fixed. I have spelled out Personal Consumption Expenditures (PCE) at first use.

The first sentence of Section A.3.1 needs revision. Likewise, the last sentence of the "Cash-only consumers" paragraph on the same page is incorrect and needs fixing.

Reply: I have revised the first sentence of the "Defining Payment Preferences" subsection in Appendix A to clarify the purpose of the data cleaning steps. I have also corrected the last sentence of the "Cash-only consumers" paragraph.

The first sentence of the proof of Theorem 1 (page A-24) requires revision.

Reply: I have revised the first sentence of the proof of Theorem 1 (the quasi-profit approximation theorem) to fix the grammar issue.

In Figure A.6, the two lines appear indistinguishable. Are they always perfectly overlapping?

Reply: The lines are very close at observed fee levels, and the difference becomes visible only for fees substantially higher than current equilibrium.

In Section C.4.0, the paper should explicitly name the figure being described.

Reply: I have added an explicit reference to the figure of the upper envelope in Figure OE.3.

In references, "et al." is fine in-text, but please list all coauthors in the bibliography.

Reply: I have verified that the bibliography lists all coauthors for every reference. The “et al.” abbreviation appears only in the in-text citations (as is standard for papers with many authors), while the bibliography entries include the complete author lists.